EXPERIMENT #3

Timestamp: 20260317_093539 | Environment: Local

1. Experiment Parameters & Network Statistics

Parameter	Value
Number of sticks (N)	30
Stick length (a)	0.7
Domain size	1.0 × 1.0
Source region	x < 0.1
Drain region	x > 0.9
Total intersections	38
Has percolation	✓ Yes
# Percolating paths	1
Nodes before pruning	36
Nodes after pruning	23
Edges before pruning	35
Edges after pruning	22

2. Network Visualizations

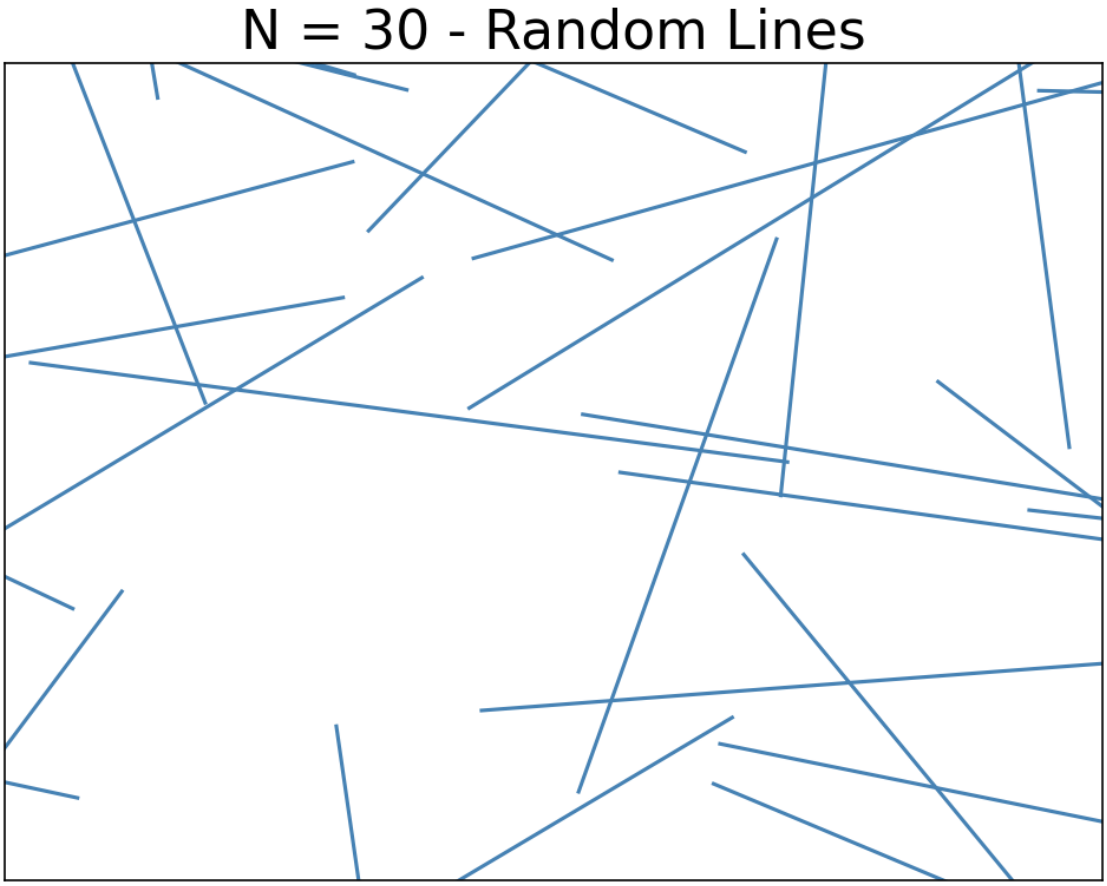


Fig. — 01 random lines N30

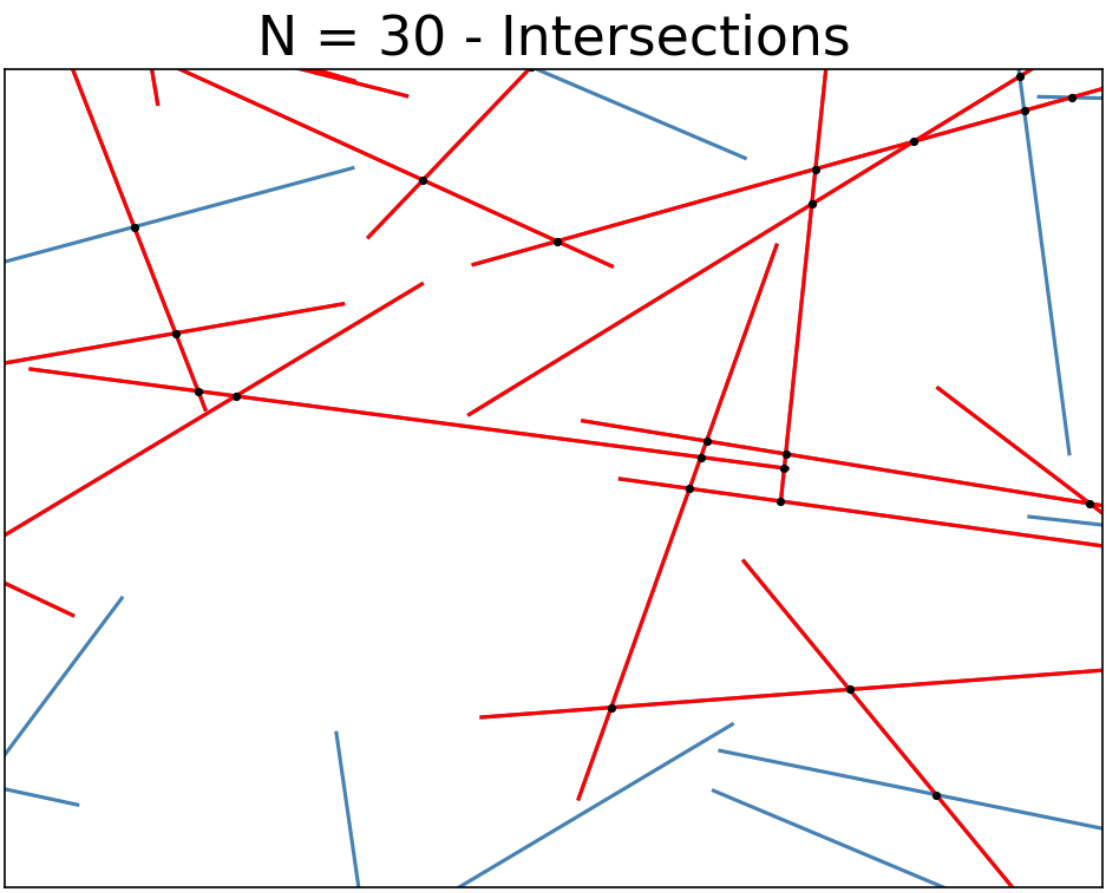


Fig. — 02 intersections N30

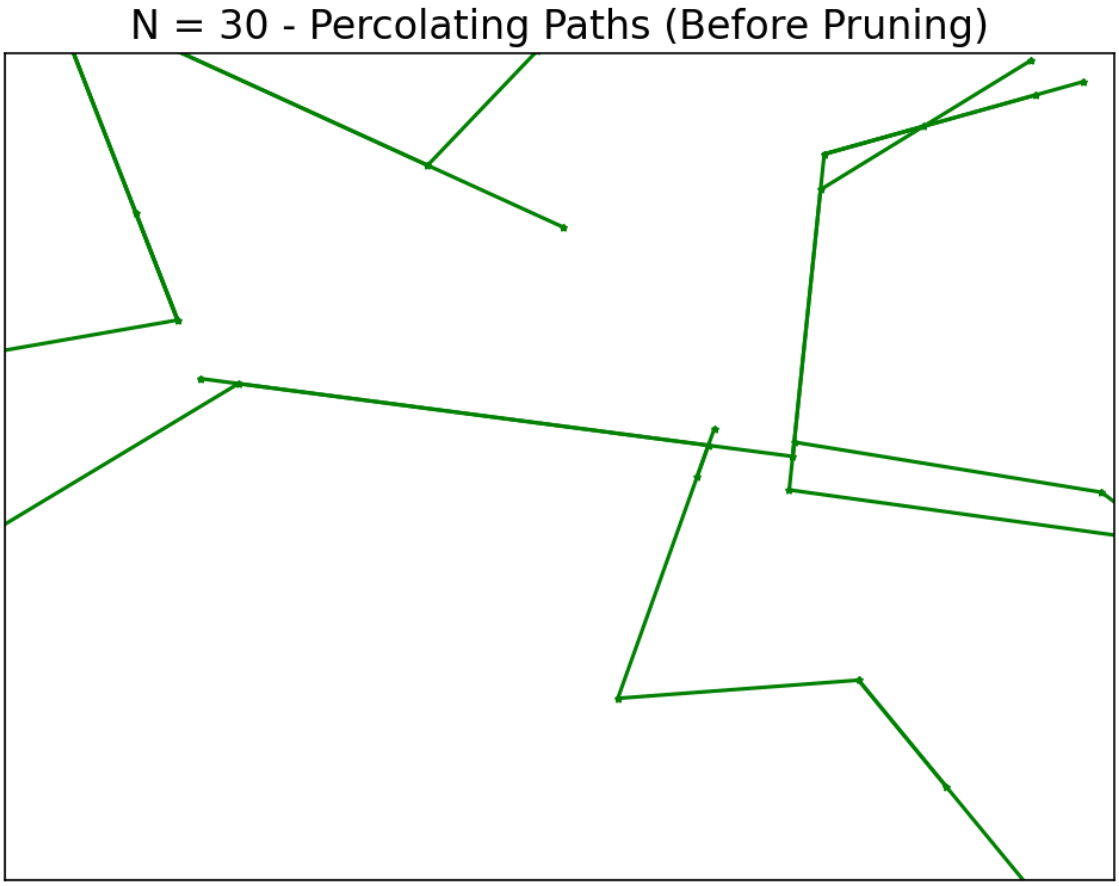


Fig. — 03 percolating paths before pruning N30

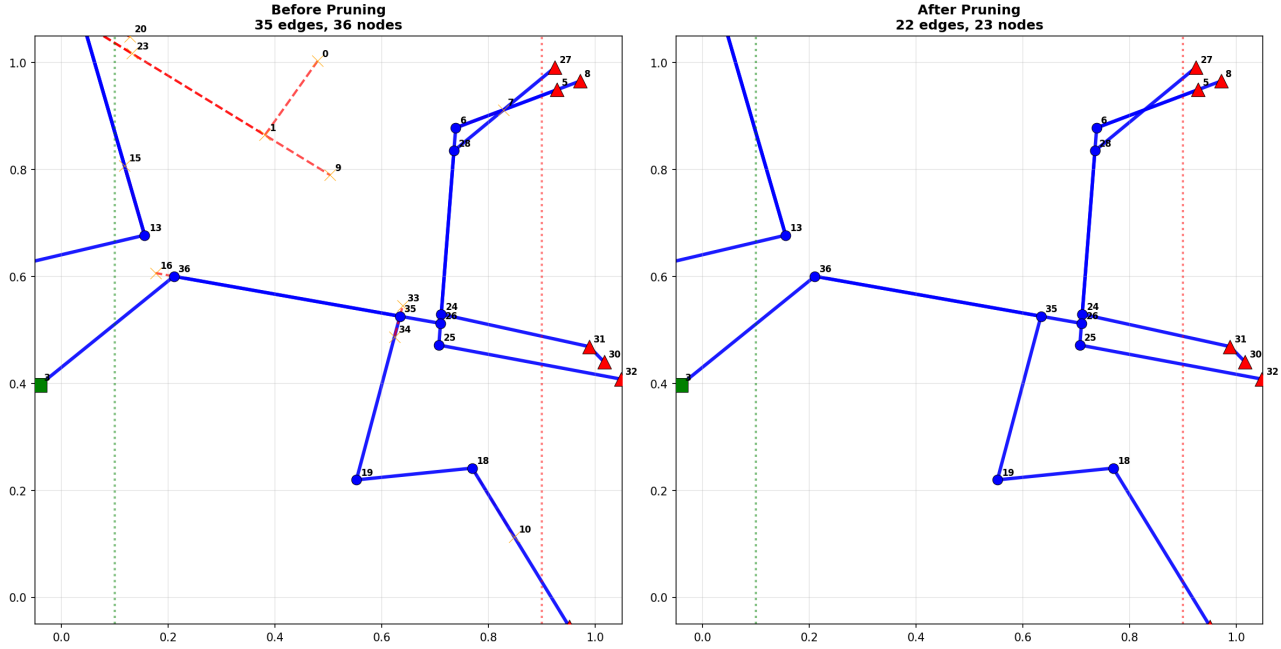


Fig. — 04 before after pruning N30

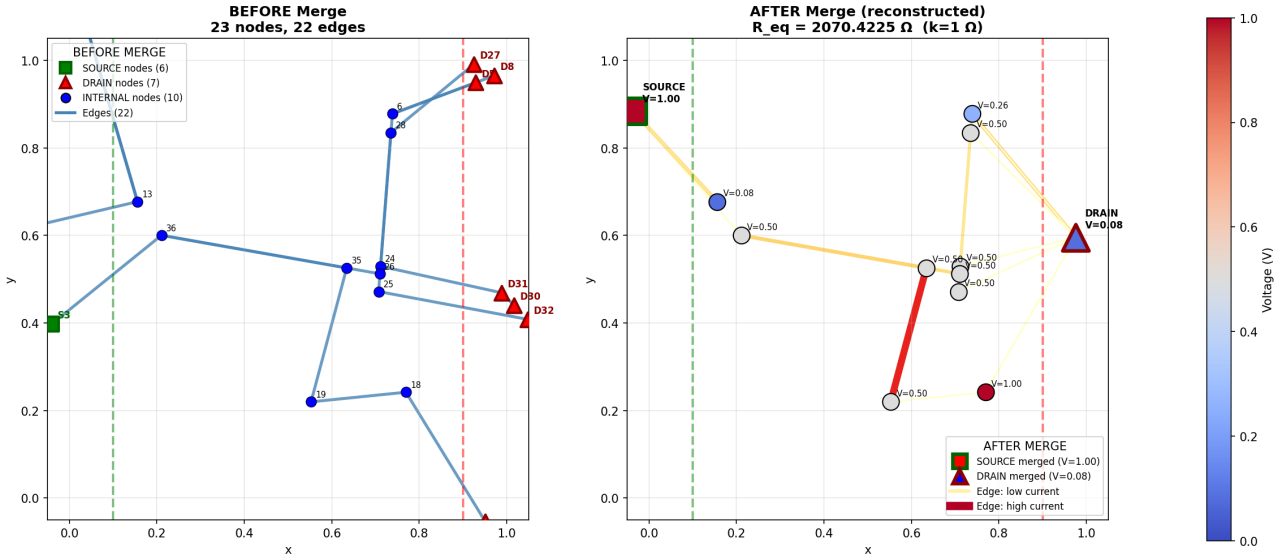


Fig. — 05 before after merge N30

3. Edge Resistances (Pruned Network)

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
0	3	4	1.0000	1.0000	0.176871
1	3	36	1.0000	1.0000	0.258503
2	4	13	1.0000	1.0000	0.081633
3	36	26	1.0000	1.0000	0.000000
4	36	35	1.0000	1.0000	0.000000
5	13	12	1.0000	1.0000	0.000000
6	13	14	1.0000	1.0000	0.000000
7	13	17	1.0000	1.0000	0.000000
8	26	6	1.0000	1.0000	0.163265
9	26	24	1.0000	1.0000	0.000000
10	26	25	1.0000	1.0000	0.000000
11	26	28	1.0000	1.0000	0.000000
12	35	19	1.0000	1.0000	0.000000
13	12	37	1.0000	1.0000	0.000000
14	6	5	1.0000	1.0000	0.081633
15	6	8	1.0000	1.0000	0.224490
16	24	31	1.0000	1.0000	0.000000
17	25	32	1.0000	1.0000	0.000000
18	28	27	1.0000	1.0000	0.000000
19	19	18	1.0000	1.0000	0.000000
20	31	30	1.0000	1.0000	0.000000
21	18	11	1.0000	1.0000	0.000000

4. Impedance Calculation

Quantity	Value
Applied voltage (V)	1.00 V
Resistance per segment k	1 Ω
Calculated impedance R_eq	2070.422535 Ω
Total current I	-1.292517e-04 A

Impedance for different k (resistance per segment) values

k (Ω/segment)	R_equivalent (Ω)
1	2.070423
10	20.704225
100	207.042254
1000	2070.422535
10000	20704.225352